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Notifications 

1. Linear algebra

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Part B due Jan 24, 2024 20:30 IST

Reduced row echelon form and rank

2/2 points (graded)

Find the reduced row echelon form of the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{pmatrix}.$$

(Enter the matrix as a list of lists. That is enter each row between square brackets, and separate rows by commas. Surround the entire matrix by square brackets; e.g. type `[[1, 0, 0], [0, 1, 0], [0, 0, 1]]` for the 3×3 identity matrix. Note when typed correctly it will render as a matrix beneath the answer box!)

rref (**A**) =

✓

Answer: `[[1, 2, 0, 3, 0, 7], [0, 0, 1, 0, 0, 1], [0, 0, 0, 0, 1, 2]]`

How many pivot columns does **rrefA** have?

✓ Answer: 3

Solution:

Either using MATLAB Online, or by performing a series of row operations, we find that

$$\begin{pmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 4 & -10 & 6 & 12 & 28 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 5 & 0 & -17 & -29 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 2 & -5 & 3 & 6 & 14 \\ 0 & 0 & -2 & 0 & 7 & 12 \\ 0 & 0 & 0 & 0 & 0.5 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & -5 & 3 & 0 & 2 \\ 0 & 0 & -2 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 2 & 0 & 3 & 0 & 7 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix}.$$

The number of pivots in **rref** (**A**) is 3.

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You have used 1 of 4 attempts

i Answers are displayed within the problem

Find the dimension of the nullspace 1

2/2 points (graded)

Find the dimension of the nullspace of the matrix

$$\mathbf{B} = \begin{pmatrix} -3 & 6 & -1 & 1 & 7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix}.$$

Calculator

$\dim(\text{NS}(\mathbf{B})) =$ ✓ Answer: 2

What the number of pivots in $\mathbf{rrefB}$?

✓ Answer: 3

Solution:

To find the dimension of nullspace of $\mathbf{B}$ and its rank, we find a row echelon form:

$$\begin{pmatrix} -3 & 6 & -1 & 1 & 7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 1/3 & -1/3 & -7/3 \\ 0 & 0 & 1 & 2 & 4/5 \\ 0 & 0 & 1 & 2 & 2/13 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

This row echelon form has 5 columns: 3 pivot columns and 2 non-pivot columns. Therefore the dimension of the nullspace is 2.

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You have used 1 of 4 attempts

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Find the dimension of the nullspace 2

2/2 points (graded)

Find a the dimension of the nullspace of the matrix

$$\mathbf{C} = \begin{pmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

$\dim(\text{NS}(\mathbf{C})) =$ ✓ Answer: 2

What is the number of pivots of $\mathbf{rrefC}$?

✓ Answer: 3

Solution:

We start by finding a row echelon form by hand or by computer:

$$\begin{pmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 3 & 0 & 3 \\ 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

This row echelon form of $\mathbf{C}$ has 3 pivot columns and 2 non-pivot columns. Therefore the dimension of the nullspace is 2.

In MATLAB, plug in matrix, apply $\mathbf{rref}$. Count the number of pivot columns, and the number of non-pivot columns, which is equal to the dimension of the nullspace.

